

FFT ASIC Design & Implementation

ADI Summer 26' Digital IC Design Internship Final Project
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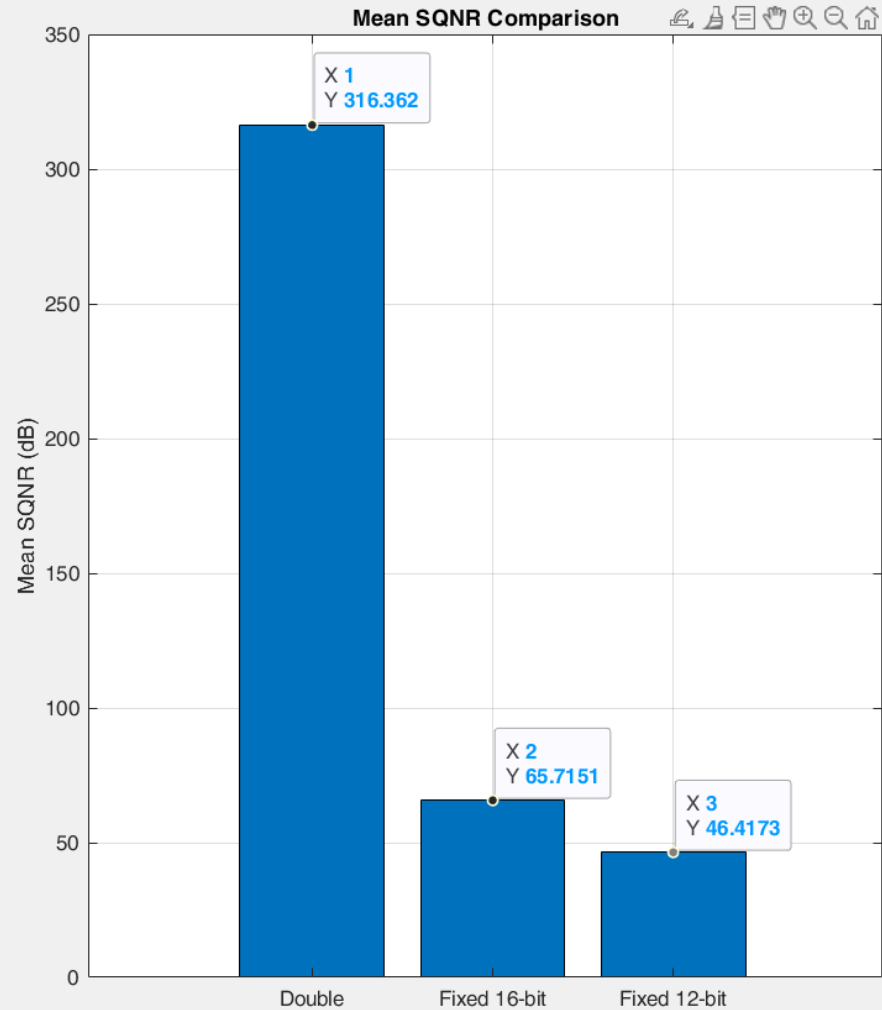
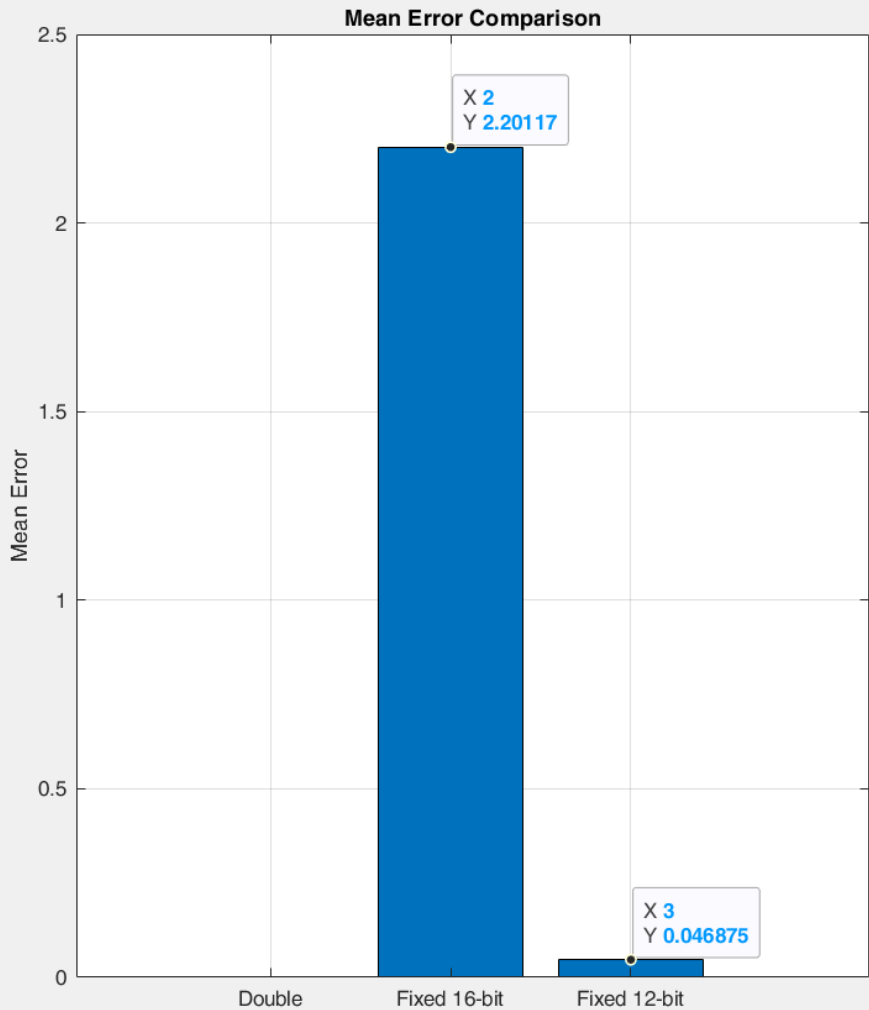
- ▶ System Modeling
- ▶ RTL Design
- ▶ RTL Verification
- ▶ ASIC Implementation
- ▶ FPGA Implementation

System Modeling

Phase 1

- ▶ Algorithms : DIT VS DIF
- ▶ Implemented 16-point Radix-2 DIF FFT Using Single-path Delay Feedback (SDF) architecture
- ▶ Initially , we model 16-point DIF algorithm using Floating Point representation
- ▶ Developed fixed-point model with optimized bit-width allocation for each stage
- ▶ Compared fixed-point implementation with floating-point implementation
- ▶ Metrics of Comparison : SQNR & Error

Results Comparison



```
===== Double Precision =====  
Mean Error : 1.870823e-15  
Max Error  : 5.024296e-15  
Mean SQNR  : 316.36 dB  
===== Fixed-Point (16-bit) =====  
Mean Error : 2.201172e+00  
Max Error  : 1.600195e+01  
Mean SQNR  : 65.72 dB  
===== Fixed-Point (12-bit) =====  
Mean Error : 4.687500e-02  
Max Error  : 7.812500e-02  
Mean SQNR  : 46.42 dB
```

- We choose 12-bit Fixed-point implementation with different integer and fraction bits for each stage

- ▶ **Number of FFT Points:** 16
- ▶ **Architecture:** DIF R2 SDF
- ▶ **Input order:** Natural order
- ▶ **Output order:** Bit-reversed order
- ▶ **Implementation:** 4 stages
- ▶ **Twiddle Factors:** Pre-computed and quantized to fixed-point representation (12-bit)

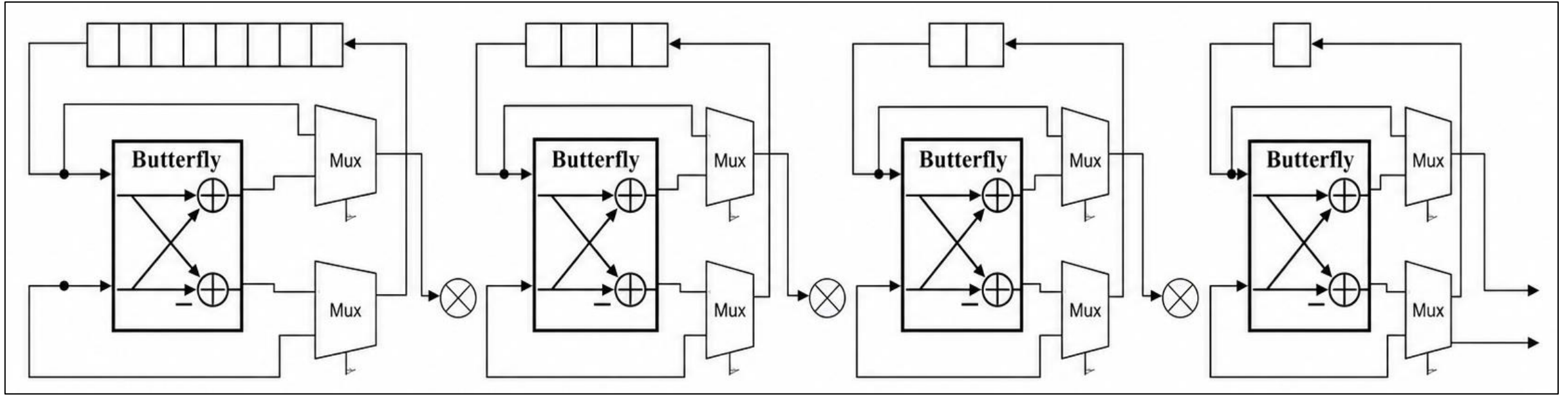
► **Word length:** 12-bit

► **Bit-Width Allocation :**

- Input signals: 4 integer + 8 fraction bits (Q4.8)
- Twiddle Factors: 2 integer + 10 fraction bits (Q2.10)
- Stage 1: 5 integer + 7 fraction bits (Q5.7)
- Stage 2: 5 integer + 7 fraction bits (Q5.7)
- Stage 3: 5 integer + 7 fraction bits (Q5.7)
- Stage 4: 6 integer + 6 fraction bits (Q6.6)
- Final Output: 6 integer + 6 fraction bits (Q6.6)

► **Rounding Method :** Flooring

SDF Architecture Implementation



► Features:

- Sequential processing
- Optimized memory usage with feedback paths.
- Butterfly operations followed by twiddle factor multiplication.
- Delay elements halve in size with each stage ($8 \rightarrow 4 \rightarrow 2 \rightarrow 1$).

RTL Design

Phase 2

► **Modular Design:**

- Four cascaded pipeline stages
- Each stage implements one level of the FFT butterfly operations
- Parameterized bit-widths for flexible precision control

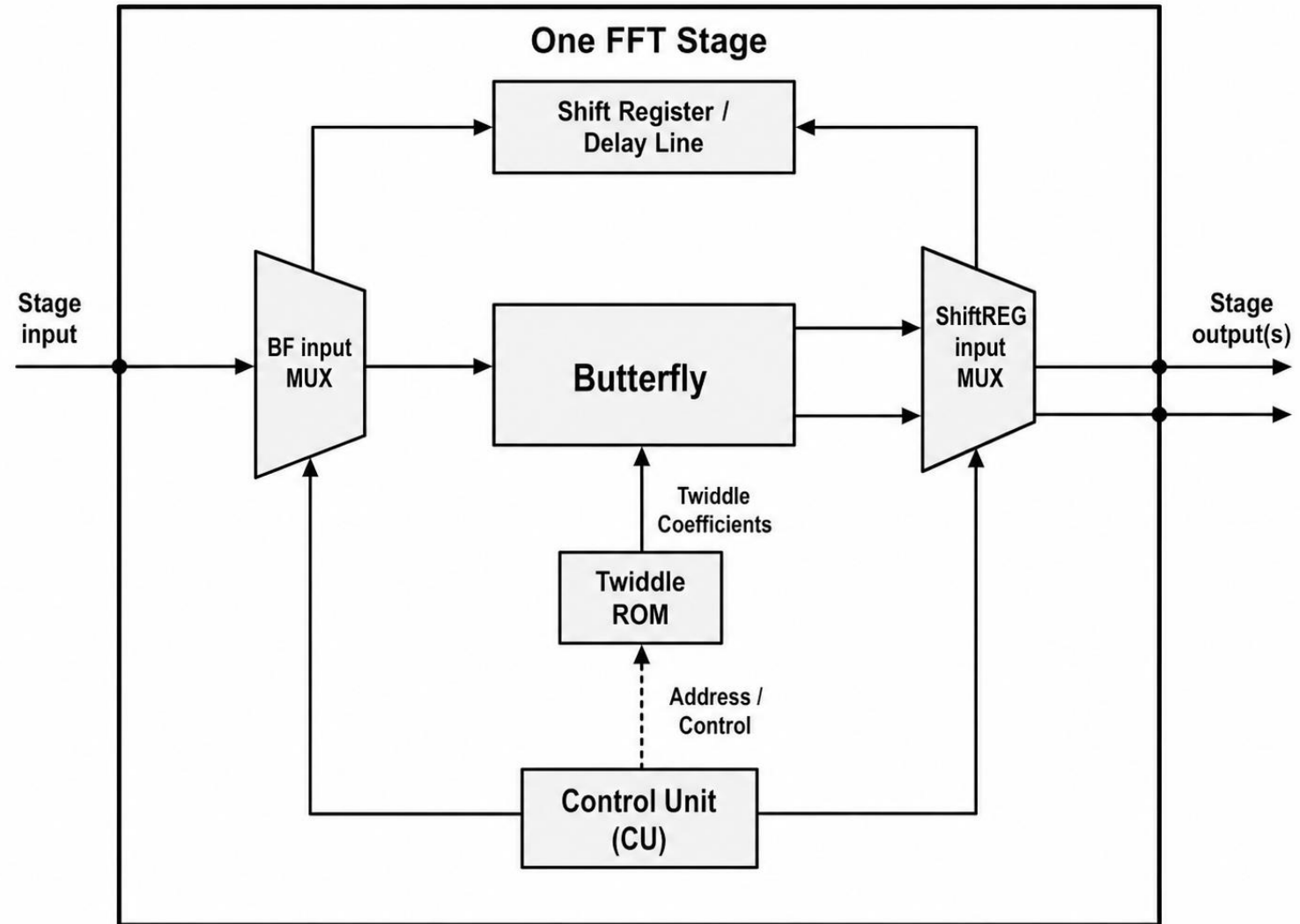
► **Control Signals:**

- First stage has invalid input signal to indicate that there is an input data
- Each stage sends valid signal to the next stage to start its operation
- Final stage has outvalid signal asserted when the output is ready

Stage Architecture

► Common stage architecture:

- Shift Register with stage-specific length
- Butterfly unit for addition /subtraction operations and instantiate complex multiplier
- ROM contain Twiddle Factor
- MUXs for input of the butterfly and the input of the shift register
- CU for MUXs selections and send addresses for the ROM



► Latency:

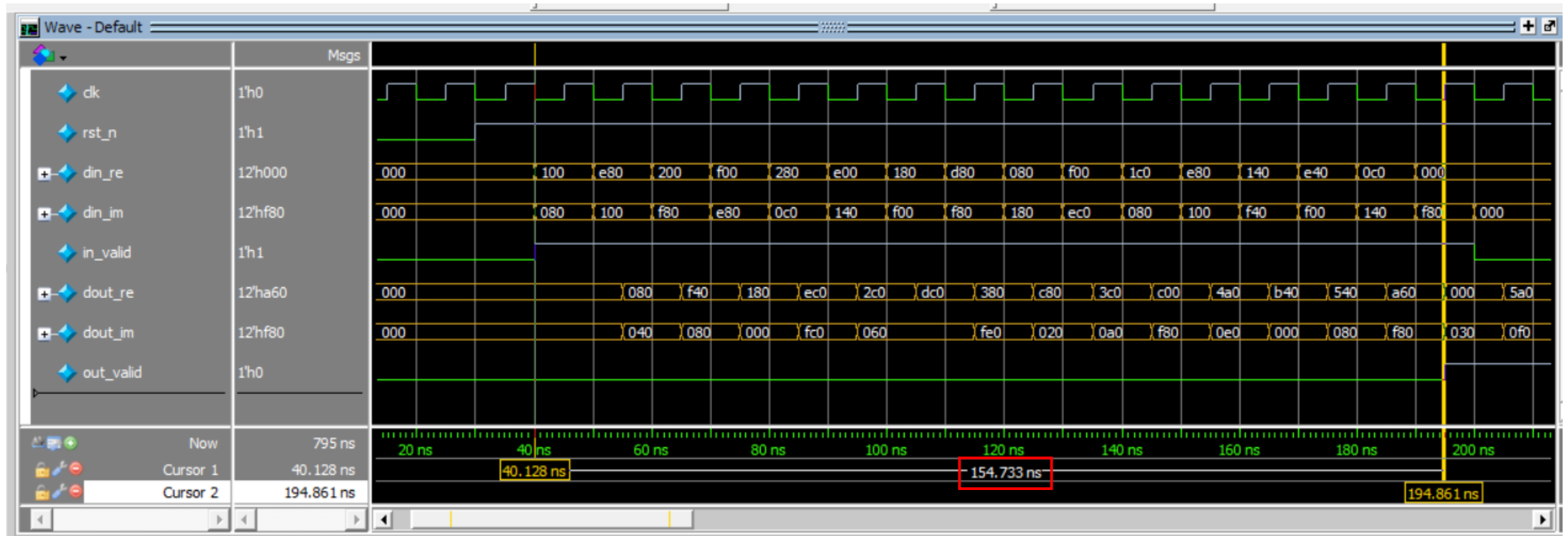
- Stage 1 : 8 cycles
- Stage 2 : 4 cycles
- Stage 3 : 2 cycles
- Stage 4 : 1 cycle
- Total Latency : 15 cycle

► Throughput:

- 1 complex sample per clock cycle (after first sample) due to pipelined architecture (SDF)

Top Module Latency

- Total Latency = 8 + 4 + 2 + 1 = 15 cycles



- 15 cycle each cycle take 10ns so total latency = 150ns

RTL Verification

Phase 3

- ▶ Comprehensive verification strategy using UVM.
- ▶ Based on end-to-end verification of complete FFT pipeline.
- ▶ Checking using either assertions or Scoreboard.
 - Scoreboard: checked the output of the data frame against expected output
 - Assertions: checked the asynchronous reset functionality and the out_valid signal also the latency (15cycle).
- ▶ Exact bit checking with MATLAB
- ▶ Generated 3 sequences to test
 - Reset seq: assert the reset signal to check reset functionality
 - Specific seq: check the behavior by entering DC signal then Impulse
 - Main seq: check that the inputs used in modeling phase in MATLAB gives the same output of the MATLAB
- ▶ Checked the **code coverage** and also made **functional coverage** to check that some sequences are happened and bins are hated or if its illegal it is not happened
- ▶ Results of the coverage represented on automated website to make reading results easier.
- ▶ Check linting the design for static RTL verification before syntesis

**ANALOG
DEVICES**

AHEAD OF WHAT'S POSSIBLE™

-
- ```
graph LR; MATLAB[MATLAB] --> TEXT[/TEXT file
contain the input and the output
corresponding this input/]; TEXT --> MainSequence((Main Sequence)); TEXT --> Scoreboard((Scoreboard));
```
- The diagram illustrates the data flow process. It begins with a blue rounded rectangle labeled "MATLAB". An arrow points from this rectangle to a green parallelogram labeled "TEXT file contain the input and the output corresponding this input". From the right side of the parallelogram, two arrows branch out to two black ovals: "Main Sequence" (top) and "Scoreboard" (bottom).



## ► UVM Report:

```
=====
 FFT VERIFICATION REPORT
=====

Total correct samples : 16032

Data mismatches : 0

MATLAB FFT frames : 1000

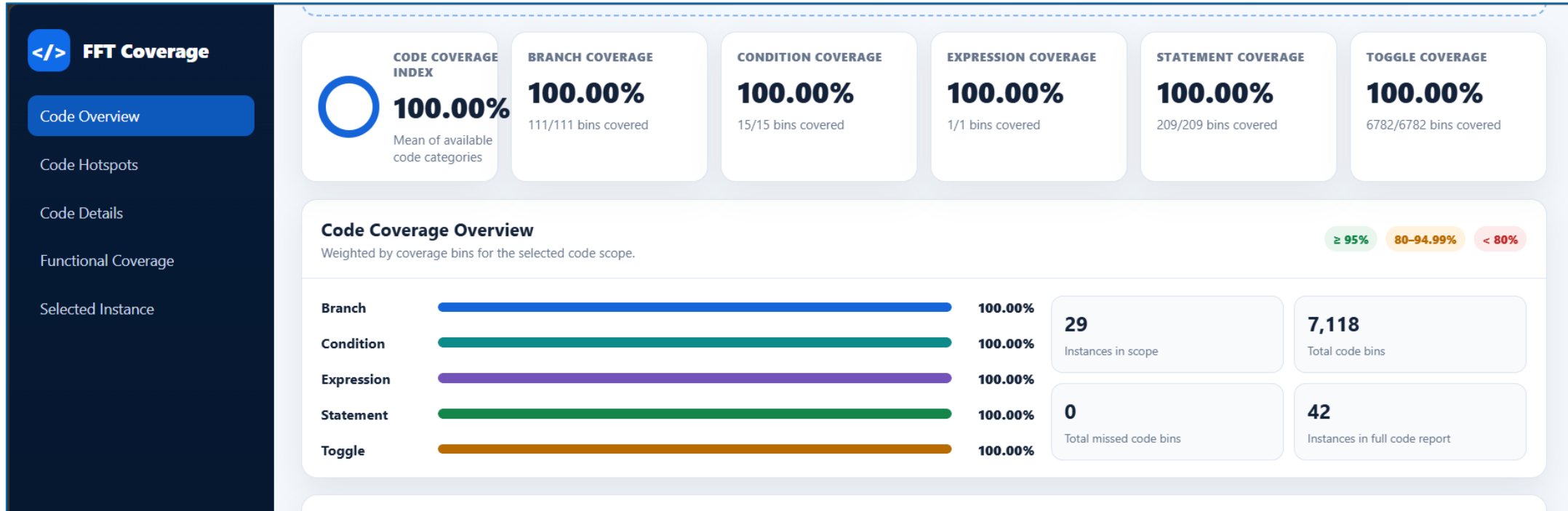
Directed tests : DC, Impulse

=====

RESULT : PASS

=====
```

## ► Code Coverage:



- Some branches like default and signal in toggle are excluded like twiddle factor signals as they are not meant to be toggled

► Functional Coverage:

TOTAL FUNCTIONAL COVERAGE

100.00%

1 covergroup type(s)

COVERPOINTS / CROSSES

7 / 2

Parsed from covergroup instances

COVERED FUNCTIONAL BINS

44/44

100.00%

IGNORE / ILLEGAL BINS

2 / 8

No illegal bins hit

/FFT\_coverage\_pkg::FFT\_coverage::cvg\_group

Covergroup instance

100.00% Covered

|                                                     |         |                 |           |         |      |
|-----------------------------------------------------|---------|-----------------|-----------|---------|------|
| <div>Reset_bins</div> <div>Coverpoint</div>         | 100.00% | 2 / 2 covered   | 0 missing | Covered | Bins |
| <div>Valid_data</div> <div>Coverpoint</div>         | 100.00% | 2 / 2 covered   | 0 missing | Covered | Bins |
| <div>Output_Validation</div> <div>Coverpoint</div>  | 100.00% | 2 / 2 covered   | 0 missing | Covered | Bins |
| <div>Real_IN</div> <div>Coverpoint</div>            | 100.00% | 5 / 5 covered   | 0 missing | Covered | Bins |
| <div>IMAG_IN</div> <div>Coverpoint</div>            | 100.00% | 3 / 3 covered   | 0 missing | Covered | Bins |
| <div>Real_out</div> <div>Coverpoint</div>           | 100.00% | 5 / 5 covered   | 0 missing | Covered | Bins |
| <div>IMAG_out</div> <div>Coverpoint</div>           | 100.00% | 3 / 3 covered   | 0 missing | Covered | Bins |
| <div>In_valid_with_data_in</div> <div>Cross</div>   | 100.00% | 11 / 11 covered | 0 missing | Covered | Bins |
| <div>Out_valid_with_data_out</div> <div>Cross</div> | 100.00% | 11 / 11 covered | 0 missing | Covered | Bins |

## ► Linting with spyglass:

**Run Goal:** lint/lint\_rtl

**Instances:** DIF\_R2SDF\_top, STAGE1, STAGE2, STAGE3, STAGE4

```

1 module DIF_R2SDF_top #(
2 parameter DATA_WIDTH = 12
3) (
4 input wire clk,
5 input wire rst_n,
6 // FFT INPUT
7 input wire signed [DATA_WIDTH-1:0] din_re,
8 input wire signed [DATA_WIDTH-1:0] din_im,
9 input wire in_valid,
10 //serial FFT OUTPUT
11 output wire signed [DATA_WIDTH-1:0] dout_re,
12 output wire signed [DATA_WIDTH-1:0] dout_im,
13 // Out valid

```

**Help Viewer:**

**Goal : lint/lint\_rtl**  
Checks the design for basic Connectivity issues  
Checks the design for basic Simulation issues  
Checks the design for recommended design practices and Structural issues  
Checks the design for basic Synthesis issues

**Description:**  
This goal checks basic connectivity issues in the design, such as floating input, width mismatch, etc. These checks should be run after every change in RTL code prior to code check-in.  
This goal checks simulation issues in the design, such as  
- incomplete sensitivity list  
- incorrect use of blocking/ non-blocking assignments  
- potential functional errors  
- possible simulation hang cases, and  
- simulation race cases  
These checks should be run, and reported messages should be reviewed prior to all simulation runs.  
This goal identifies the structural issues in the design that affect the

**Group By:** Severity

| Message                                                                                                           | File                                                                  | Line |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------|
| Message Tree ( Total: 2, Displayed: 2, Waived: 0)                                                                 |                                                                       |      |
| INFO (2)                                                                                                          |                                                                       |      |
| DetectTopDesignUnits (1) : Identify the top-level design units in user design.                                    |                                                                       |      |
| Module DIF_R2SDF_top is a top level design unit                                                                   | rtl/top.v                                                             | 1    |
| ElabSummary (1) : Generates Elaborated design units Summary data                                                  |                                                                       |      |
| Please refer file './spyglass-1/lint/lint_rtl/spyglass_reports/SpyGlass/elab_summary.rpt' for elab summary report | ./spyglass-1/lint/lint_rtl/spyglass_reports/SpyGlass/elab_summary.rpt | 0    |

**Violations:** Shell, Violations, Waiver Tree

**Search:** Search in Message Tree

# ASIC Flow

## Phase 4

- ▶ **Technology:** SkyWater130nm (sky130\_fd\_sc\_hd -High Density library)
- ▶ **Target:** reach frequency of at least 50MHz with min area as possible
- ▶ **Constraint on time:**
  - Uncertainty for hold 0.2ns
  - Uncertainty for setup 0.1ns
  - Period 20ns
- ▶ **Area Constraints:**
  - Core utilization in placement 25% "due to high congestion"
  - Density 50%
- ▶ **Flow done:**
  - Synthesis -> Formality -> floorplan -> powerplan -> placement -> CTS -> Routing -> chip finishing -> signoff verification

## Area report

```
1
2 =====
3 report_design_area
4 =====
5 Design area 124218 u^2 100% utilization.
```

## Worst Slack report

```
1 |
2 =====
3 report_worst_slack -max (Setup)
4 =====
5 worst slack 12.32
6
7 =====
8 report_worst_slack -min (Hold)
9 =====
10 worst slack 0.19
```



## Input Files Required For Formality Check

Design Verilog Files (.v)

Netlist From synthesis (.v)

Guidance File (.svf)

Std cells libraries (.lib/.db)

## Failing Point Report

```

Report : failing_points

Reference : Ref:/WORK/DIF_R2SDF_top
Implementation : Imp:/WORK/DIF_R2SDF_top
Version : L-2016.03-SP1
Date : Fri Aug 14 00:02:22 2026

No failing compare points.

1
```

## Unverified Points Report

```

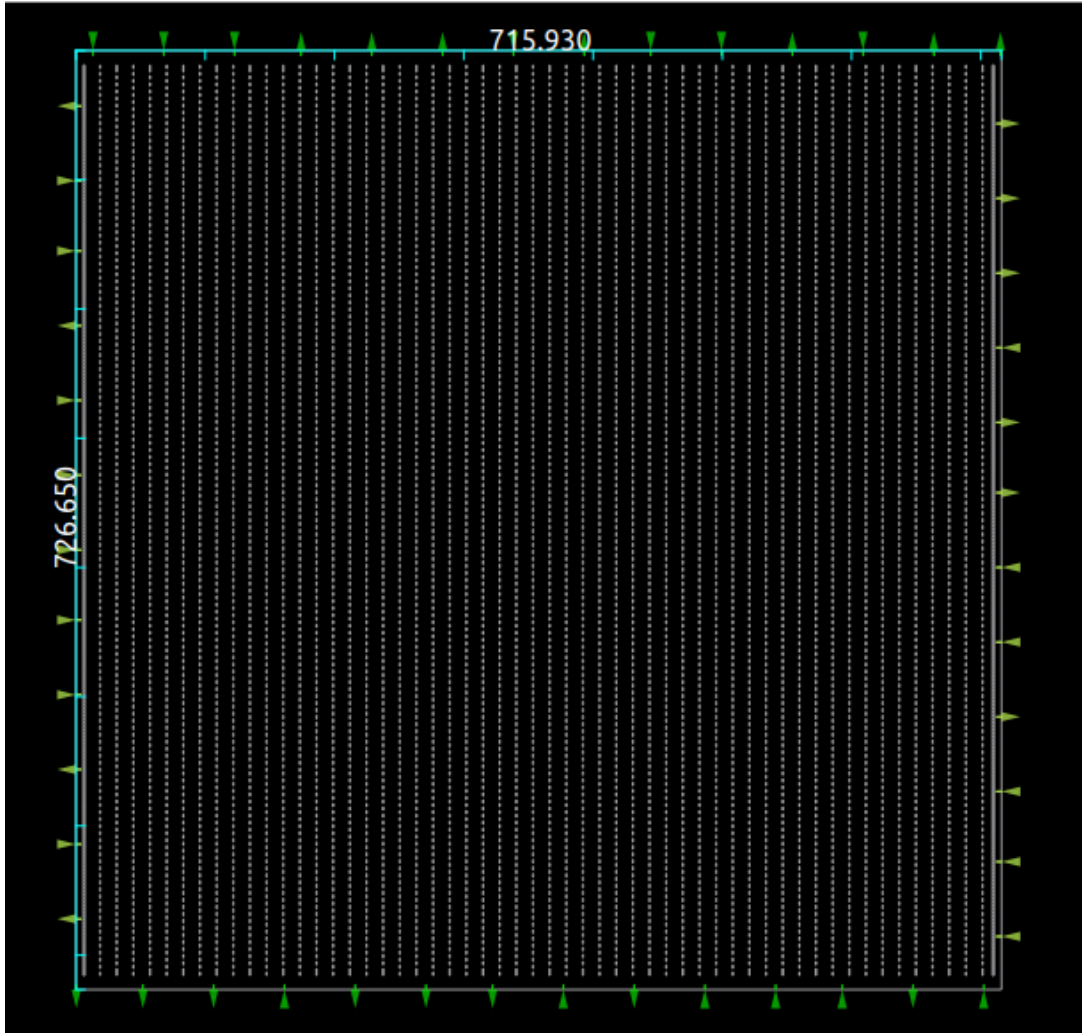
Report : unverified_points

Reference : Ref:/WORK/DIF_R2SDF_top
Implementation : Imp:/WORK/DIF_R2SDF_top
Version : L-2016.03-SP1
Date : Fri Aug 14 00:02:22 2026

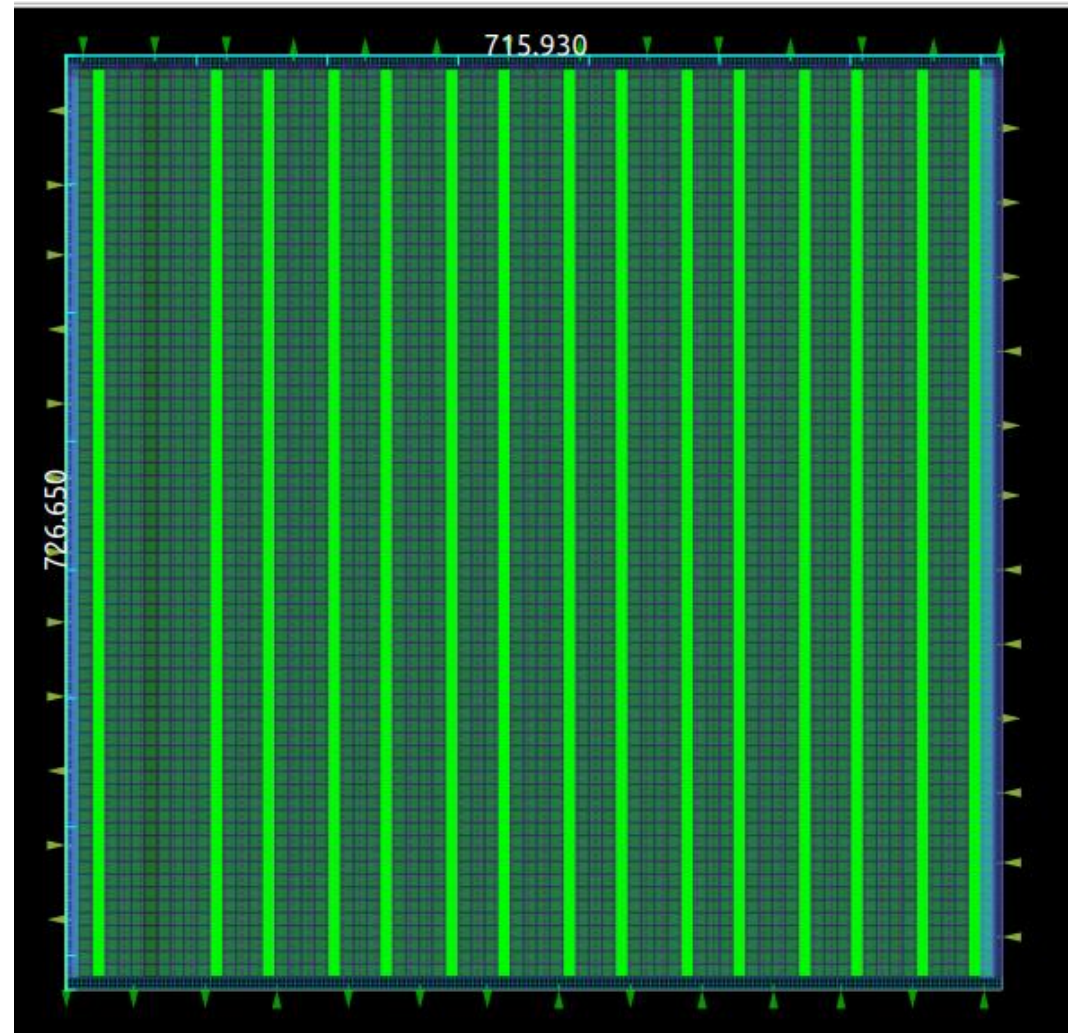
No unverified compare points.

1
```

Without metal layers view

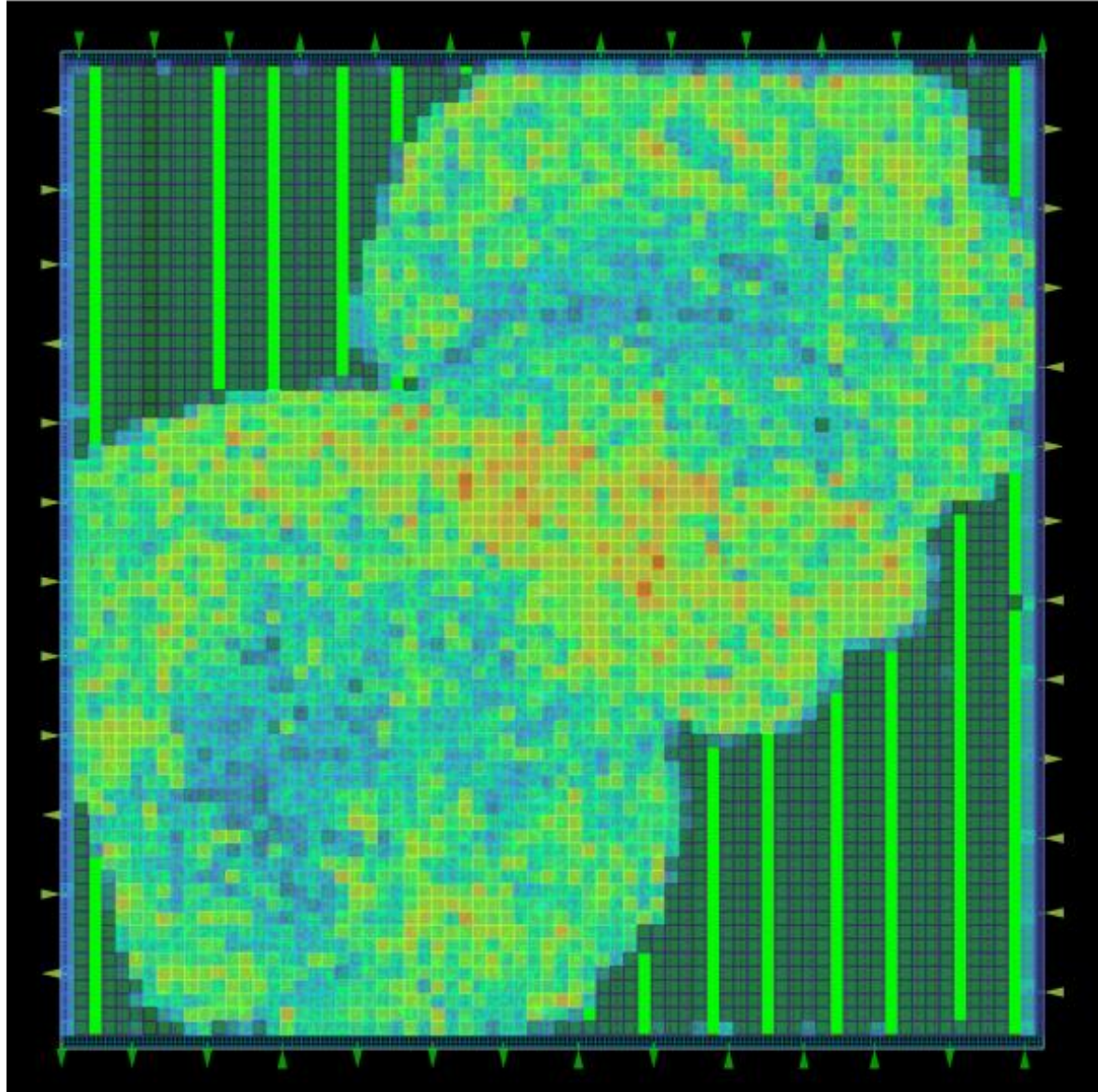


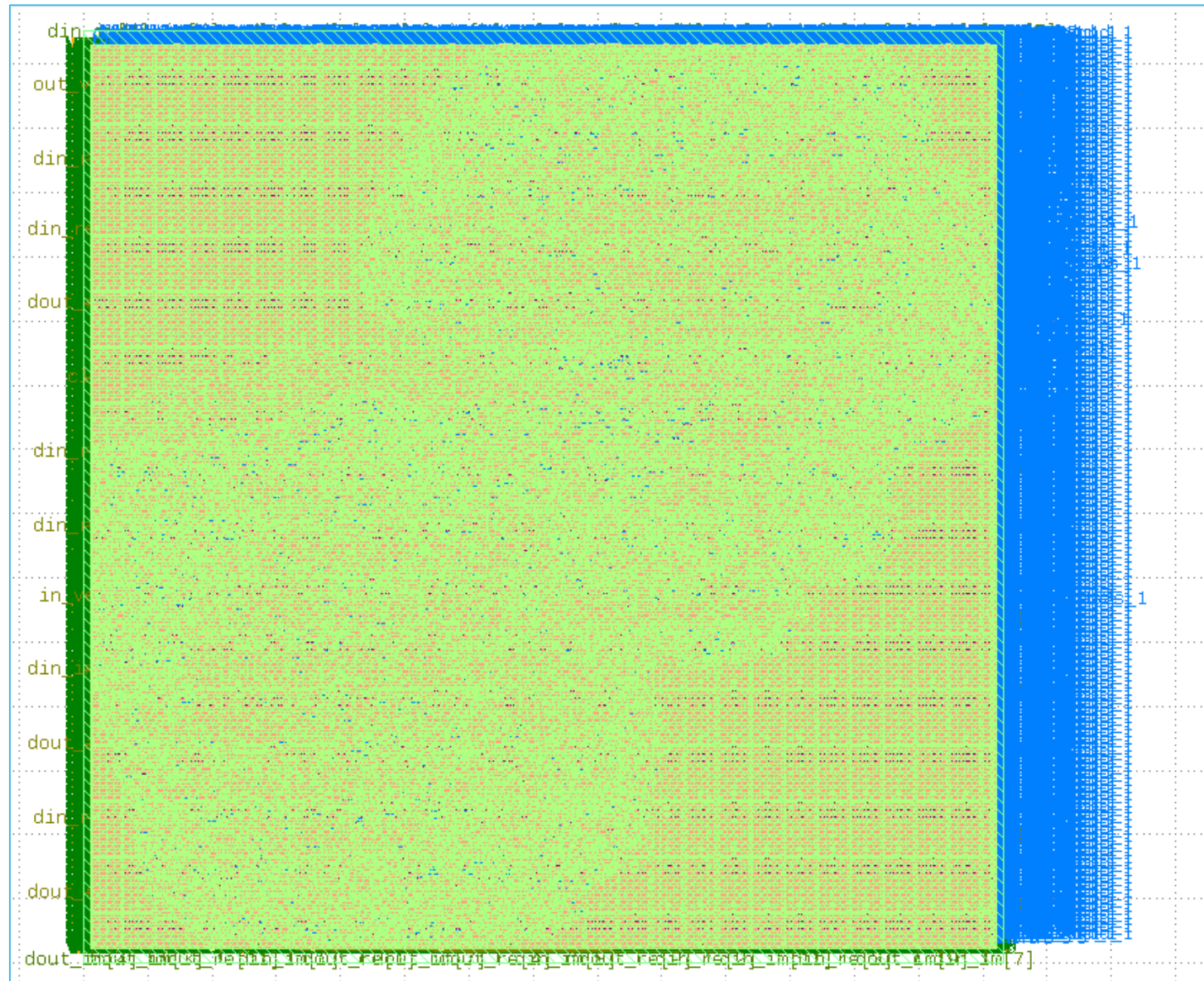
With metal layers view





# Placement





| Metrics                                               | Value                                |
|-------------------------------------------------------|--------------------------------------|
| Worst Negative Slack (WNS)<br>for Setup               | 4.48ns                               |
| Worst Negative Slack (WNS)<br>for Hold                | 0.01ns                               |
| Total Negative Slack (TNS)<br>for both setup and hold | 0ns                                  |
| Max Frequency                                         | $\frac{1}{(20 - 4.48)ns} = 64.43MHz$ |
| Area & Utilization                                    | 483583μm <sup>2</sup> & 97%          |

```
1
2 =====
3 report_design_area
4 =====
5 Design area 483583 u^2 97% utilization.
```

```
1 |
2 =====
3 report_worst_slack -max (Setup)
4 =====
5 worst slack 4.48
6
7 =====
8 report_worst_slack -min (Hold)
9 =====
10 worst slack 0.01
```

```
1 |
2 =====
3 report_wns
4 =====
5 wns 0.00
```

```
1
2 =====
3 report_tns
4 =====
5 tns 0.00
```

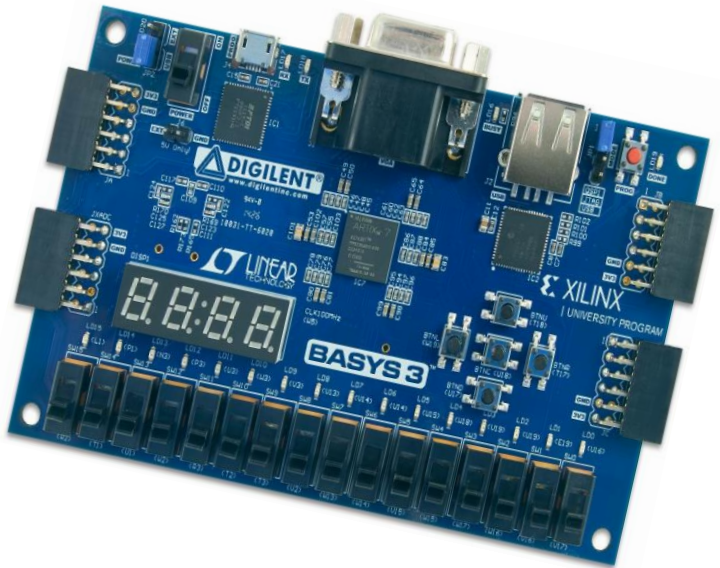


# FPGA Flow

## Phase 5

## ► Board used:

- Digilent Basys 3 board
- ARTIX-7 FPGA

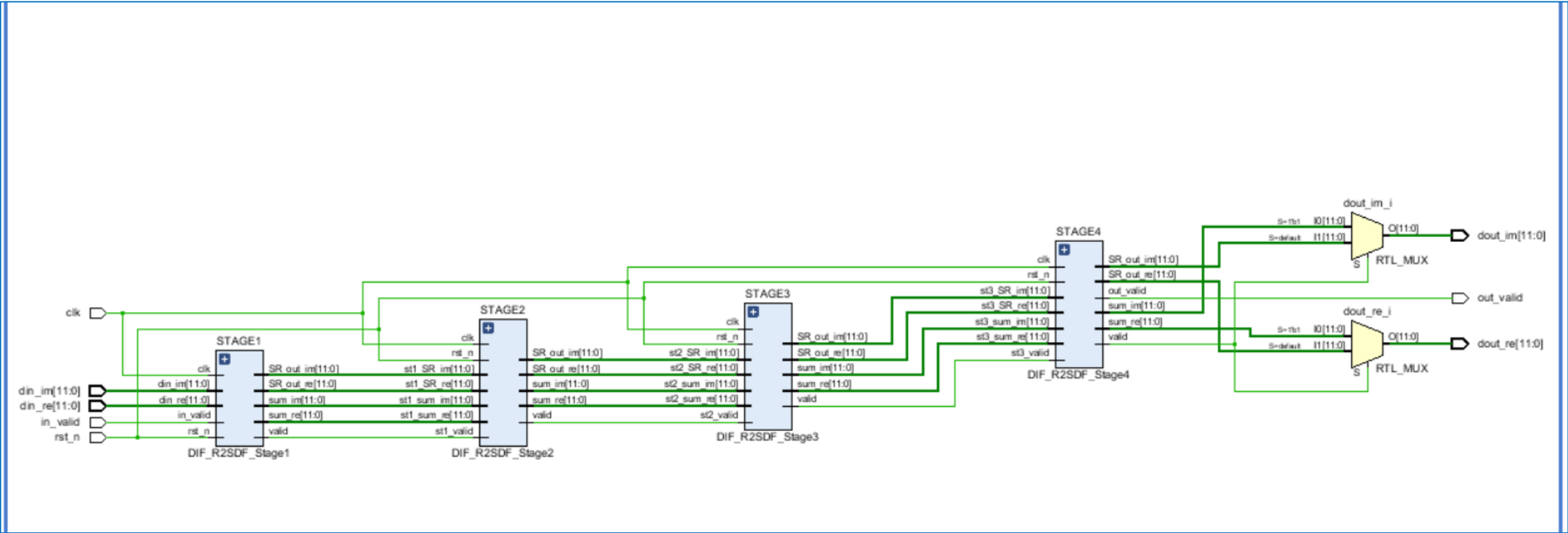


## ► Main constraints used:

- Clk period = 20nsec

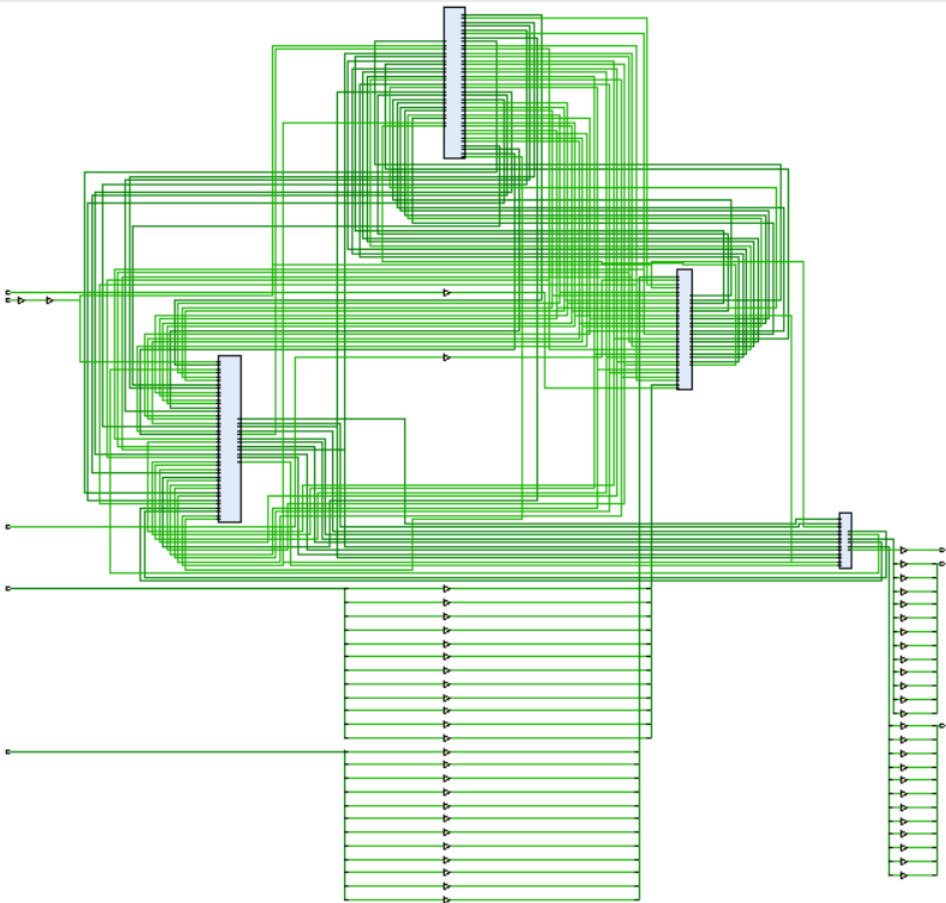
```
1 ## -----
2 ## 50 MHz SYSTEM CLOCK
3 ## -----
4
5 set_property PACKAGE_PIN W5 [get_ports clk]
6 set_property IOSTANDARD LVCMOS33 [get_ports clk]
7 create_clock -add -name sys_clk_pin -period 20.000 -waveform {0.000 10.000} [get_ports clk]
8
9
10 ## -----
```

► Schematic:





## ► Schematic:



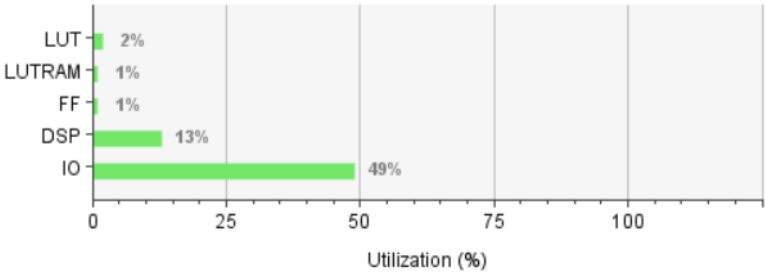
## ► Utilization & Timing:

Design Timing Summary

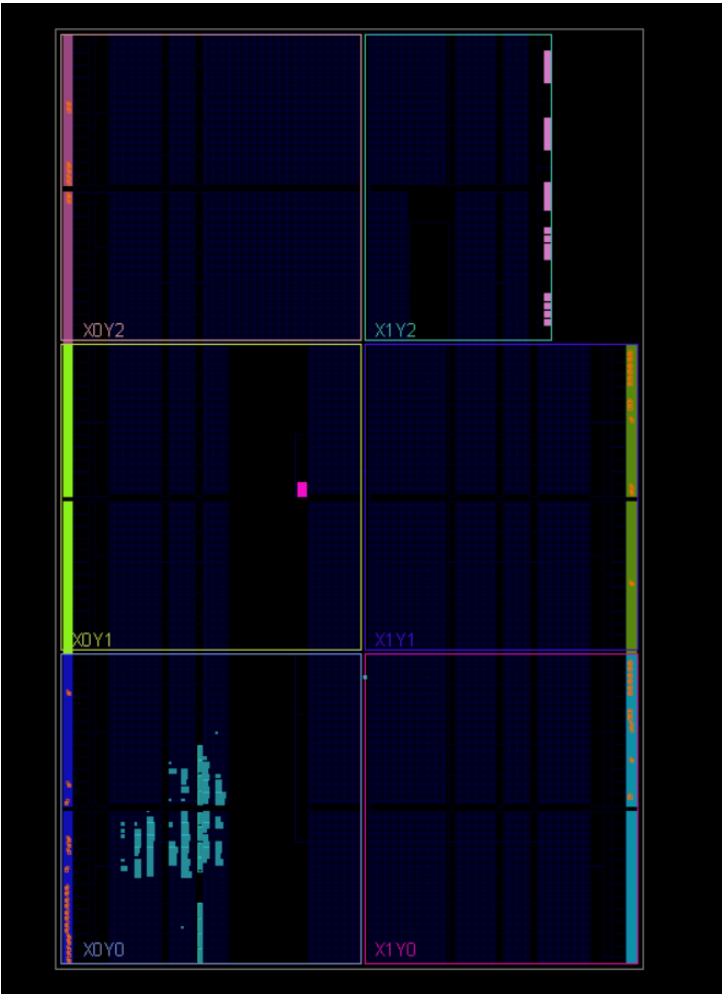
| Setup                                | Hold                             | Pulse Width                                       |
|--------------------------------------|----------------------------------|---------------------------------------------------|
| Worst Negative Slack (WNS): 6.219 ns | Worst Hold Slack (WHS): 0.139 ns | Worst Pulse Width Slack (WPWS): 9.020 ns          |
| Total Negative Slack (TNS): 0.000 ns | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0       | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 269       | Total Number of Endpoints: 269   | Total Number of Endpoints: 276                    |

All user specified timing constraints are met.

| Resource | Utilization | Available | Utilization % |
|----------|-------------|-----------|---------------|
| LUT      | 399         | 20800     | 1.92          |
| LUTRAM   | 48          | 9600      | 0.50          |
| FF       | 227         | 41600     | 0.55          |
| DSP      | 12          | 90        | 13.33         |
| IO       | 52          | 106       | 49.06         |



► Device:



► Utilization & Timing:

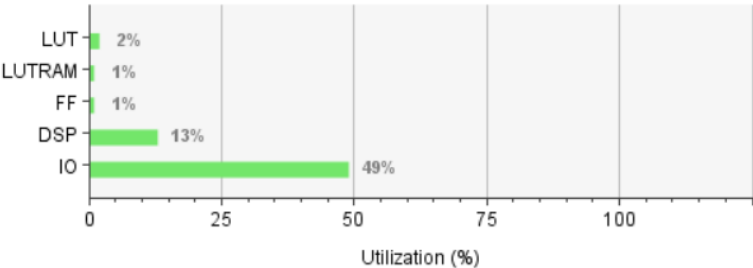
Design Timing Summary

| Setup                                | Hold                             | Pulse Width                                       |
|--------------------------------------|----------------------------------|---------------------------------------------------|
| Worst Negative Slack (WNS): 3.770 ns | Worst Hold Slack (WHS): 0.126 ns | Worst Pulse Width Slack (WPWS): 9.020 ns          |
| Total Negative Slack (TNS): 0.000 ns | Total Hold Slack (THS): 0.000 ns | Total Pulse Width Negative Slack (TPWS): 0.000 ns |
| Number of Failing Endpoints: 0       | Number of Failing Endpoints: 0   | Number of Failing Endpoints: 0                    |
| Total Number of Endpoints: 269       | Total Number of Endpoints: 269   | Total Number of Endpoints: 276                    |

All user specified timing constraints are met.

Summary

| Resource | Utilization | Available | Utilization % |
|----------|-------------|-----------|---------------|
| LUT      | 399         | 20800     | 1.92          |
| LUTRAM   | 48          | 9600      | 0.50          |
| FF       | 227         | 41600     | 0.55          |
| DSP      | 12          | 90        | 13.33         |
| IO       | 52          | 106       | 49.06         |

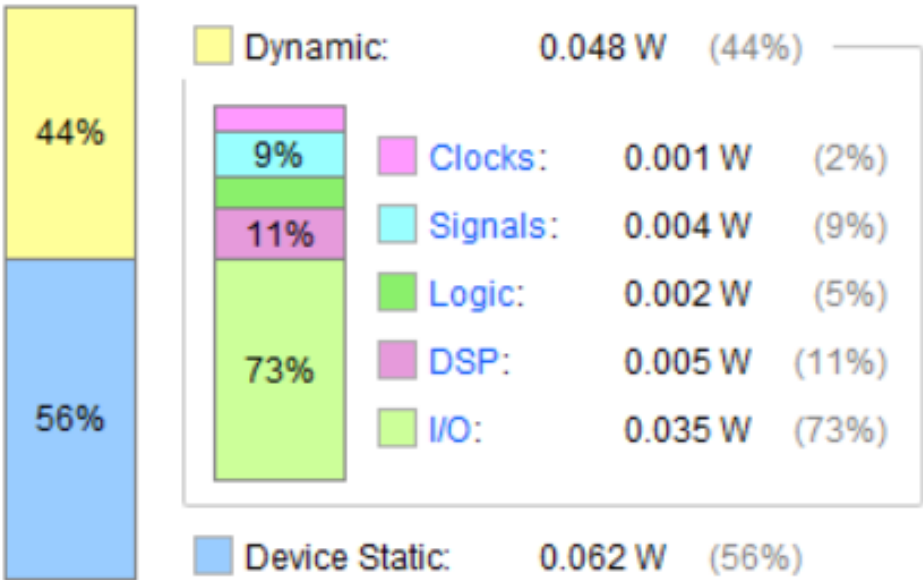


Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

|                                     |                 |
|-------------------------------------|-----------------|
| Total On-Chip Power:                | 0.11 W          |
| Design Power Budget:                | Not Specified   |
| Power Budget Margin:                | N/A             |
| Junction Temperature:               | 25.5°C          |
| Thermal Margin:                     | 74.5°C (14.8 W) |
| Effective $\theta_{JA}$ :           | 5.0°C/W         |
| Power supplied to off-chip devices: | 0 W             |
| Confidence level:                   | Low             |

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



# Thank You!